

# C+L Band Tunable External Cavity Laser Array Integrated using Photonic Wire Bonding

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**Abstract** We report on an 8-channel external cavity-based TLS integrated using PWB between a RSOA array and a SiN circuit array. We demonstrate a wide tunability over C+L band, output power >16 dBm, SMSR >60 dB and intrinsic linewidth <10 KHz. ©2026 The Author(s)

## Introduction

External cavity diode laser (ECDL) arrays have attracted much attention in many applications as a cost-effective solution. In particular, in applications where wavelength division multiplexing (WDM) is needed to meet the escalating capacity demands of modern data centers, ECDL array-based modules are an attractive approach, allowing a smaller footprint, wide tunability, narrow linewidths, high power, and reduced assembly costs compared to multiple tunable laser source (TLS) modules [1, 2].

State-of-the-art multi-wavelength sources often rely on arrays of distributed feedback (DFB) lasers [3-5]. For instance, recent high-performance modules utilize 200 GHz-spaced DFB arrays to generate multiple carriers (16 channels) [3]. However, the fixed-wavelength nature of DFBs limits their flexibility in dynamic optical networks and wavelength routing. To address this, TLS arrays, particularly ECDL arrays, are highly desirable to provide wide, flexible wavelength coverage [1].

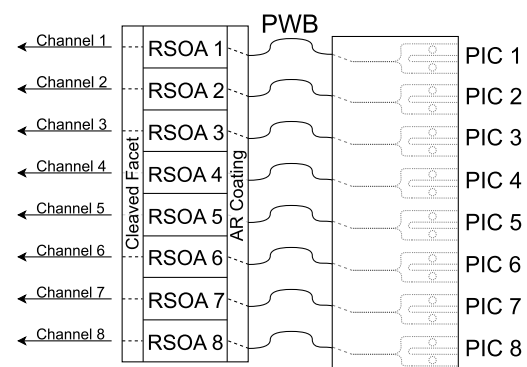
The adoption of photonic integrated circuit (PIC)-based ECDL arrays is, however, subject to some issues when relying on traditional hybrid integration techniques: stringent sub-micrometer active alignment tolerances for flip-chip bonding, and spatial alignment requirements across multiple axes alongside optical mode mismatches for butt-coupling. These challenges complicate the assembly process, limit yield when scaled to multi-channel arrays, and degrade the performance of the laser by inducing coupling loss and intra-cavity feedback [4-7].

In this paper, we report on an 8-channel external cavity-based TLS integrated using Photonic Wire Bonding (PWB) between a reflective semiconductor optical amplifier (RSOA)

array and a single-chip SiN PIC array. We first describe the components and the integration process. We then present the characterization results of the TLS array, detailing the L-I curves, output power, side-mode suppression ratio (SMSR), and the Lorentzian linewidth across the C+L band.

## TLS Array Description

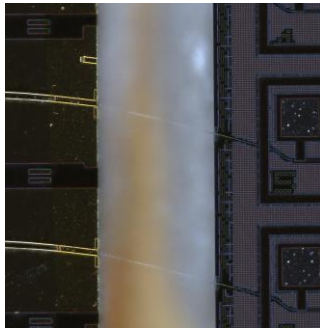
Optical gain in the TLS is provided by a monolithic RSOA array featuring a quantum well active region within a buried heterostructure. The epitaxial structure was designed to achieve a photoluminescence peak that enables coverage of the C+L band. The RSOA back facet, which serves as the laser output, was cleaved, while the front facet coupled to the PIC was anti-reflection coated. In order to suppress unwanted optical feedback, the RSOA stripes were angled at 7° relative to the cleaved facet.



**Fig.1:** Schematic of the 8-channel TLS array.

The external cavity is based on a silicon nitride PIC (fabricated in a LIGENTEC AN800 multi-project wafer run) that employs Vernier effect-based tunable reflectors. Each of the 8 circuits feature two Micro-Ring Resonators (MRR) in a serial add-drop configuration (Manhattan

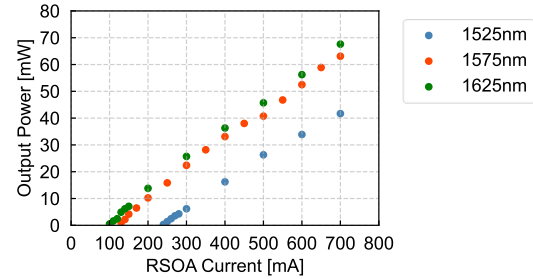
configuration). The MRRs are designed to yield a free spectral range (FSR) that exceeds 100 nm across the C+L band, and also to allow a trade-off between power loss and filter selectivity. Coupling between each RSOA and SiN circuit is achieved using PWBs, which are fabricated via two-photon polymerization in a negative-tone photoresist (Vanguard Automation GmbH) (see Fig.2). The AR coating of the RSOA was optimized to match the PWB refractive index. A low refractive index cladding was then used to protect the PWBs [7].



**Fig.2:** Picture of the PWB connecting RSOA (Left) and PIC (Right) before the cladding deposition.

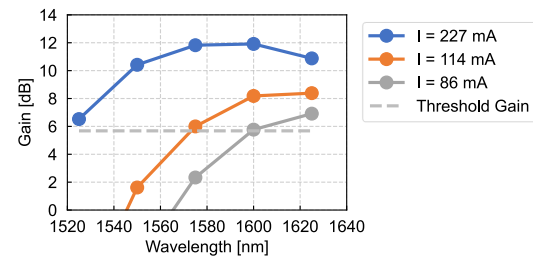
#### TLS Array Characterization: L-I curves, Output Power and SMSR

We first evaluated the L-I curves, output power, SMSR, and spectral tuning characteristics. The RSOA was driven with injection currents up to 700 mA., while its temperature was monitored by a negative temperature coefficient (NTC) thermistor and stabilized at 25 °C using a thermoelectric controller (TEC). External cavity wavelength tuning was thermo-electrically controlled using heaters: one MRR handled coarse tuning, while a phase section fine-tuned the alignment of the cavity mode with the Vernier filter main peak, thereby maximizing the output power. All reported power measurements represent fiber-coupled power, obtained by collecting light from the RSOA facet via a lens and coupling it into an optical fiber. Fig.3 illustrates the L-I curves of the 5th channel in the TLS array, obtained by sweeping the RSOA bias current from 700 mA down to the threshold current for three wavelengths across the C+L band. A peak output power of 16 dBm (~40 mW) was recorded at 1525 nm, and 18 dBm (~63 mW) at 1575 nm and 1625 nm. The extracted threshold current values were 227 mA, 114 mA, and 86 mA ( $\pm 8$  mA) at 1525 nm, 1575 nm, and 1625 nm respectively. The higher threshold current observed at 1525 nm is primarily due to the lower



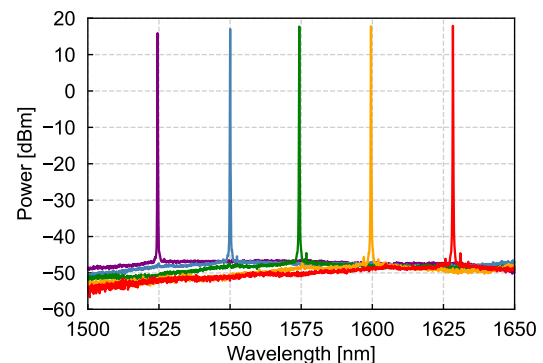
**Fig.3:** L-I curves of the 5th channel in the TLS array.

gain at this wavelength (Fig. 4). To extract the PWB coupling loss, we first simulated the RSOA gain spectra for each measured threshold current, yielding a threshold gain of ~5.8 dB. Because gain equals total cavity loss at the lasing threshold, this value represents our total cavity loss (Fig. 4). By subsequently subtracting the known SiN filter, RSOA internal, and facet losses from this total, we isolated the PWB coupling loss, revealing a consistent value of  $2.5 \text{ dB} \pm 0.5 \text{ dB}$  across the tuning range.



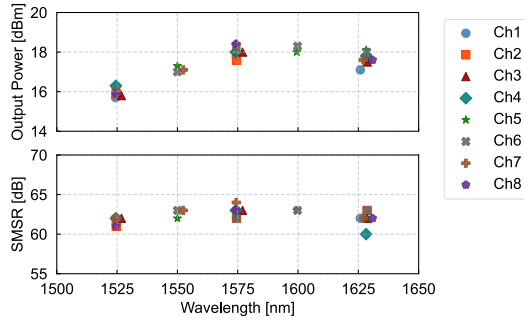
**Fig.4:** Simulated RSOA single-pass gain vs wavelength at the measured threshold current at 1525nm, 1575nm and 1625nm.

The wavelength tuning characteristics of the TLS array were evaluated at a constant RSOA bias current of 700 mA. A wavelength sweep was conducted to capture the lasing spectra across the tuning range. As a representative example, Fig.5 plots the spectra of the fifth TLS channel for five distinct wavelengths across the C+L band,



**Fig.5:** Spectra across the tuning range of the 5th channel in the TLS array.

demonstrating stable single-mode lasing over a 100 nm span with an SMSR larger than 60 dB. Subsequently, the output power and SMSR were extracted for multiple TLS channels across the tuning range, as shown in Fig.6. The array demonstrated uniform performance, with maximum inter-channel variations limited to ~1 dB for output power and ~3 dB for SMSR.

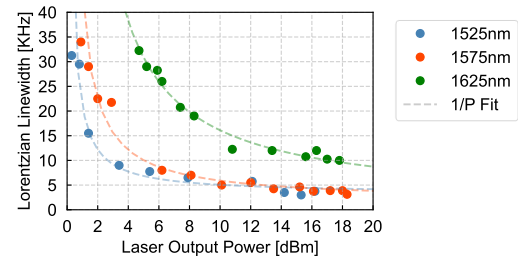


**Fig.6:** Measured output power and SMSR vs wavelength of multiple channels in the TLS array.

### TLS Array Characterization: Intrinsic Linewidth

To evaluate the TLS array in terms of intrinsic/Lorentzian linewidth, we adapted a heterodyne phase noise measurement method using an offline digital coherent receiver [8]. The technique involves optically beating the laser under test with a local oscillator (LO), which is a commercial TLS, and capturing the resulting signal's complex amplitude using the digital coherent receiver. The digitized data is then analysed using offline digital signal processing (DSP) to extract the time-dependent phase differences. From the beat signal, the method computes the field spectrum, the frequency modulation (FM) noise spectrum, and the phase-error variance. The intrinsic linewidth is then obtained from the floor of the FM noise spectrum. To illustrate the phase noise characteristics of the TLS array, the Lorentzian linewidth of the fifth channel was measured at three distinct wavelengths at various output powers by adjusting the RSOA injection current. These measurements are presented in Fig.7. Each reported linewidth value represents the average over multiple acquisitions.

As illustrated in Fig.7, the linewidth demonstrates a clear  $1/P$  dependence across all three wavelengths. At high output powers, the measured linewidth shows a plateau at ~3 kHz for 1525 nm and 1575 nm, and at ~10 kHz for 1625 nm. Because this heterodyne measurement relies on extracting the beat signal between the device under test and a commercial TLS acting as the local oscillator (LO), this measurement floor is fundamentally restricted by the intrinsic



**Fig.7:** Measured linewidth vs laser output power using the 5th channel in the TLS array.

linewidth of the LO itself, which may be wavelength-dependent. Consequently, the actual intrinsic linewidth of our integrated TLS array at high output powers may be lower than these recorded limits. In particular, the larger values obtained at 1625nm are probably due to a larger Linewidth Enhancement Factor of our TLS, or of the commercial TLS we have used.

### Conclusion

We demonstrated a high-performance 8-channel external cavity-based TLS array, integrated using PWB between a monolithic RSOA array and a SiN PIC. The module exhibits wide tunability over a 100 nm span in the C+L band, stable single-mode lasing with an SMSR > 60 dB, and high output powers ranging from 16 dBm to 18 dBm when operated at 25 °C. Furthermore, a comprehensive characterization including L-I curves, spectral tuning, and phase noise was conducted, demonstrating a consistent PWB coupling loss of ~2.5 dB across the tuning range and an intrinsic Lorentzian linewidth below 10 kHz. Compared to state-of-the-art multi-wavelength modules relying on fixed-frequency DFB arrays, we have demonstrated a tunable TLS array enabling flexible wavelength routing. Furthermore, delivering a relatively flat, high output power across the C+L band makes such TLS arrays an ideal fit for high-capacity WDM in modern data centers.

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